



J1 Using vectors

Name: _____

Class: _____

Date: _____

Time: **429 minutes**

Marks: **359 marks**

Comments:

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Q1.

A quadrilateral has vertices A , B , C and D with position vectors given by

$$\overrightarrow{OA} = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}, \overrightarrow{OB} = \begin{bmatrix} -1 \\ 2 \\ 7 \end{bmatrix}, \overrightarrow{OC} = \begin{bmatrix} 0 \\ 7 \\ 6 \end{bmatrix} \text{ and } \overrightarrow{OD} = \begin{bmatrix} 4 \\ 10 \\ 0 \end{bmatrix}$$

- (a) Write down the vector $\overrightarrow{AB}$

(1)

- (b) Show that $ABCD$ is a parallelogram, but not a rhombus.

(5)

(Total 6 marks)

Q2.

The three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ are acting on a particle.

$$\mathbf{F}_1 = (25\mathbf{i} + 12\mathbf{j}) \text{ N}$$

$$\mathbf{F}_2 = (-7\mathbf{i} + 5\mathbf{j}) \text{ N}$$

$$\mathbf{F}_3 = (15\mathbf{i} - 28\mathbf{j}) \text{ N}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

The resultant of these three forces is $\mathbf{F}$ newtons.

- (a) (i) Find the magnitude of $\mathbf{F}$, giving your answer to three significant figures.

(2)

- (ii) Find the acute angle that $\mathbf{F}$ makes with the horizontal, giving your answer to the nearest 0.1°

(2)

- (b) The fourth force, $\mathbf{F}_4$, is applied to the particle so that the four forces are in equilibrium.

Find $\mathbf{F}_4$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(1)

(Total 5 marks)

Q3.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally.

The velocity, $\mathbf{v} \text{ m s}^{-1}$, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that the parachutist is at the origin when $t = 0$

Model the parachutist as a particle.

- (a) Find an expression for the position vector of the parachutist at time t . (4)
- (b) The parachutist opens her parachute when she has travelled 100 metres horizontally.
- Find the vertical displacement of the parachutist from the origin when she opens her parachute. (4)
- (c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model. (3)
- (Total 11 marks)**

Q4.

In this question use $g = 9.81 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector $(40\mathbf{i} - 10\mathbf{j})$ metres.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that there are no resistance forces acting on the ball.

- (a) Find the speed of the ball when it is at a height of 3 metres above its initial position. (6)
- (b) State the speed of the ball when it is at its maximum height. (1)
- (c) Explain why the answer you found in part (b) may not be the actual speed of the ball when it is at its maximum height. (1)
- (Total 8 marks)**

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Q5.

A particle, of mass 10 kg, moves on a smooth horizontal plane. At time t seconds, the acceleration of the particle is given by

$$\{(40t + 3t^2)\mathbf{i} + 20e^{-4t}\mathbf{j}\} \text{ m s}^{-2}$$

where the vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

- (a) At time $t = 1$, the velocity of the particle is $(6\mathbf{i} - 5e^{-4}\mathbf{j}) \text{ m s}^{-1}$.
- Find the velocity of the particle at time t . (5)
- (b) Calculate the initial speed of the particle. (3)

(Total 8 marks)

Q6.

A particle, of mass 50 kg, moves on a smooth horizontal plane. A single horizontal force

$$[(300t - 60t^2)\mathbf{i} + 100e^{-2t}\mathbf{j}] \text{ newtons}$$

acts on the particle at time t seconds.

The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

- (a) Find the acceleration of the particle at time t .

(2)

- (b) When $t = 0$, the velocity of the particle is $(7\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1}$.

Find the velocity of the particle at time t .

(4)

- (c) Calculate the speed of the particle when $t = 1$.

(4)

(Total 10 marks)

Q7.

A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

- (c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

Q8.

A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

- (c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

Q9.

A particle moves with constant acceleration $(-0.4\mathbf{i} + 0.2\mathbf{j}) \text{ m s}^{-2}$. Initially, it has velocity $(4\mathbf{i} + 0.5\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the particle at time t seconds.

(2)

- (b) (i) Find the velocity of the particle when $t = 22.5$.

(2)

- (ii) State the direction in which the particle is travelling at this time.

(1)

- (c) Find the time when the speed of the particle is 5 m s^{-1} .

(6)

(Total 11 marks)

Q10.

The velocity of a particle at time t seconds is $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = (4 + 3t^2)\mathbf{i} + (12 - 8t)\mathbf{j}$$

- (a) When $t = 0$, the particle is at the point with position vector $(5\mathbf{i} - 7\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(4)

- (b) Find the acceleration of the particle at time t .

(2)

- (c) The particle has mass 2 kg .

Find the magnitude of the force acting on the particle when $t = 1$.

(4)

(Total 10 marks)

Q11.

A helicopter is initially hovering above a lighthouse. It then sets off so that its acceleration is $(0.5\mathbf{i} + 0.375\mathbf{j}) \text{ m s}^{-2}$. The helicopter does not change its height above sea level as it moves. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the speed of the helicopter 20 seconds after it leaves its position above the lighthouse.

(4)

- (b) Find the bearing on which the helicopter is travelling, giving your answer to the nearest degree.

(3)

- (c) The helicopter stops accelerating when it is 500 metres from its initial position.

Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

(5)

(Total 12 marks)

Q12.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the velocity of the particle, $\mathbf{v} \text{ m s}^{-1}$, is given by

$$\mathbf{v} = 4e^{-2t} \mathbf{i} + (6t - 3t^2) \mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t .

(3)

- (b) The mass of the particle is 5 kg.

- (i) Find an expression for the force $\mathbf{F}$ acting on the particle at time t .

(2)

- (ii) Find the magnitude of $\mathbf{F}$ when $t = 0$.

(2)

- (c) Find the value of t when $\mathbf{F}$ acts due west.

(2)

- (d) When $t = 0$, the particle is at the point with position vector $(6\mathbf{i} + 5\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 14 marks)

Q13.

A boat moves with constant acceleration, so that its velocity $\mathbf{v} \text{ m s}^{-1}$ at time t seconds is given by

$$\mathbf{v} = (4.5 - 0.9t) \mathbf{i} + (2 + 0.5t) \mathbf{j}$$

where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the velocity of the boat when $t = 0$. (1)
 - (b) Find the velocity of the boat when $t = 5$. (1)
 - (c) State the direction in which the boat is travelling when $t = 5$. (1)
 - (d) Find the acceleration of the boat. (3)
 - (e) The mass of the boat is 250 kg. Find the magnitude of the resultant force acting on the boat. (3)
 - (f) Draw a diagram to show the direction of the resultant force on the boat and calculate the angle between this force and the unit vector $\mathbf{i}$. (4)
- (Total 13 marks)**

Q14.

The constant forces $\mathbf{F}_1 = (8\mathbf{i} + 12\mathbf{j})$ newtons and $\mathbf{F}_2 = (4\mathbf{i} - 4\mathbf{j})$ newtons act on a particle. No other forces act on the particle.

- (a) Find the resultant force acting on the particle. (2)
 - (b) Given that the mass of the particle is 4 kg, show that the acceleration of the particle is $(3\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. (2)
 - (c) At time t seconds, the velocity of the particle is $\mathbf{v} \text{ m s}^{-1}$.
 - (i) When $t = 20$, $\mathbf{v} = 40\mathbf{i} + 32\mathbf{j}$.
 Show that $\mathbf{v} = -20\mathbf{i} - 8\mathbf{j}$ when $t = 0$. (3)
 - (ii) Write down an expression for $\mathbf{v}$ at time t . (1)
 - (iii) Find the times when the speed of the particle is 8 m s^{-1} . (6)
- (Total 14 marks)**

Q15.

A ship moves with constant acceleration. At time t seconds, its velocity is $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Write down the velocity of the ship when $t = 0$. (1)
 - (b) Find the acceleration of the ship. (2)
 - (c) Find t when the ship is travelling south-east. (3)
- (Total 6 marks)**

Q16.

A particle, of mass 10 kg, moves on a smooth horizontal surface. A single horizontal force, $(9\mathbf{i} + 12\mathbf{j})$ newtons, acts on the particle.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the acceleration of the particle. (2)
 - (b) At time t seconds, the velocity of the particle is $\mathbf{v} \text{ m s}^{-1}$. When $t = 0$, the velocity of the particle is $(2.2\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$ and the particle is at the origin.
 - (i) Find the distance between the particle and the origin when $t = 5$. (4)
 - (ii) Express $\mathbf{v}$ in terms of t . (2)
 - (iii) Find t when the particle is travelling north-east. (3)
- (Total 11 marks)**

Q17.

A particle has mass 200 kg and moves on a smooth horizontal plane. A single

horizontal force, $\left(400 \cos\left(\frac{\pi}{2}t\right)\mathbf{i} + 600t^2\mathbf{j}\right)$ newtons, acts on the particle at time t seconds.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the acceleration of the particle at time t . (2)
- (b) When $t = 4$, the velocity of the particle is $(-3\mathbf{i} + 56\mathbf{j}) \text{ m s}^{-1}$.

Find the velocity of the particle at time t .

(5)

(c) Find t when the particle is moving due west.

(3)

(d) Find the speed of the particle when it is moving due west.

(2)

(Total 12 marks)

Q18.

Two forces, $\mathbf{P} = (6\mathbf{i} - 3\mathbf{j})$ newtons and $\mathbf{Q} = (3\mathbf{i} + 15\mathbf{j})$ newtons, act on a particle. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

(a) Find the resultant of $\mathbf{P}$ and $\mathbf{Q}$.

(2)

(b) Calculate the magnitude of the resultant of $\mathbf{P}$ and $\mathbf{Q}$.

(2)

(c) When these two forces act on the particle, it has an acceleration of $(1.5\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. Find the mass of the particle.

(2)

(d) The particle was initially at rest at the origin.

(i) Find an expression for the position vector of the particle when the forces have been applied to the particle for t seconds.

(2)

(ii) Find the distance of the particle from the origin when the forces have been applied to the particle for 2 seconds.

(2)

(Total 10 marks)

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Q19.

A particle moves on a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

At time t seconds, the position vector of the particle is $\mathbf{r}$ metres, where

$$\mathbf{r} = \left(2e^{\frac{1}{2}t} - 8t + 5 \right) \mathbf{i} + (t^2 - 6t) \mathbf{j}$$

(a) Find an expression for the velocity of the particle at time t .

(3)

(b) (i) Find the speed of the particle when $t = 3$.

(2)

(ii) State the direction in which the particle is travelling when $t = 3$.

(1)

(c) Find the acceleration of the particle when $t = 3$.

(3)

(d) The mass of the particle is 7 kg.

Find the magnitude of the resultant force on the particle when $t = 3$.

(3)

(Total 12 marks)

Q20.

A particle moves on a smooth horizontal plane. It is initially at the point A , with position vector

$(9\mathbf{i} + 7\mathbf{j})$ m, and has velocity $(-2\mathbf{i} + 2\mathbf{j})$ m s⁻¹. The particle moves with a constant acceleration of $(0.25\mathbf{i} + 0.3\mathbf{j})$ m s⁻² for 20 seconds until it reaches the point B . The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

(a) Find the velocity of the particle at the point B .

(3)

(b) Find the velocity of the particle when it is travelling due north.

(4)

(c) Find the position vector of the point B .

(3)

(d) Find the average velocity of the particle as it moves from A to B .

(2)

(Total 12 marks)

Q21.

A Jet Ski is at the origin and is travelling due north at 5 m s⁻¹ when it begins to accelerate uniformly. After accelerating for 40 seconds, it is travelling due east at 5 m s⁻¹. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

(a) Show that the acceleration of the Jet Ski is $(0.1\mathbf{i} - 0.125\mathbf{j})$ m s⁻².

(4)

(b) Find the position vector of the Jet Ski at the end of the 40 second period.

(3)

(c) The Jet Ski is travelling southeast t seconds after it leaves the origin.

(i) Find t .

(5)

(ii) Find the velocity of the Jet Ski at this time.

(2)
 (Total 14 marks)

Q22.

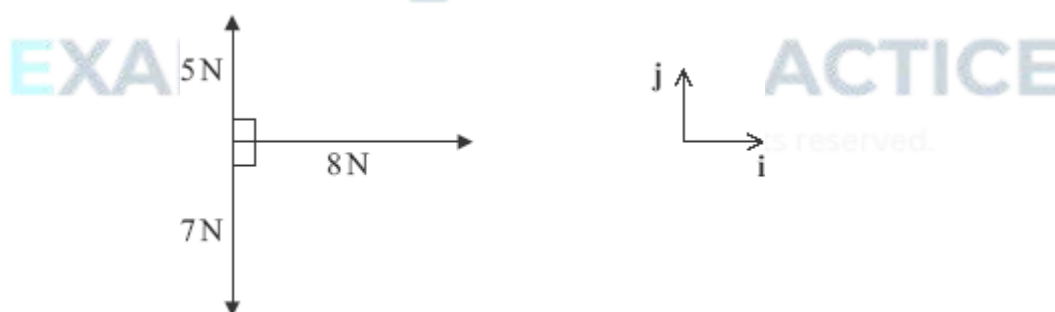
A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the position vector, $\mathbf{r}$ metres, of the particle is given by

$$\mathbf{r} = (t^3 - 3t^2 + 4)\mathbf{i} + (4t + t^2)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t . (2)
 - (b) The mass of the particle is 3 kg.
 - (i) Find an expression for $\mathbf{F}$ at time t . (3)
 - (ii) Find the magnitude of $\mathbf{F}$ when $t = 3$. (2)
 - (c) Find the value of t when $\mathbf{F}$ acts due north. (2)
- (Total 9 marks)

Q23.

The diagram shows three forces and the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$, which all lie in the same plane.



- (a) Express the resultant of the three forces in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)
 - (b) Find the magnitude of the resultant force. (2)
 - (c) Draw a diagram to show the direction of the resultant force, and find the angle that it makes with the unit vector $\mathbf{i}$. (3)
- (Total 7 marks)

Q24.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A helicopter moves horizontally with a constant acceleration of $(-0.4\mathbf{i} + 0.5\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the helicopter is at the origin and has velocity $20\mathbf{i} \text{ m s}^{-1}$.

- (a) Write down an expression for the velocity of the helicopter at time t seconds. (2)
 - (b) Find the time when the helicopter is travelling due north. (3)
 - (c) Find an expression for the position vector of the helicopter at time t seconds. (2)
 - (d) When $t = 100$:
 - (i) show that the helicopter is due north of the origin; (3)
 - (ii) find the speed of the helicopter. (3)
- (Total 13 marks)**

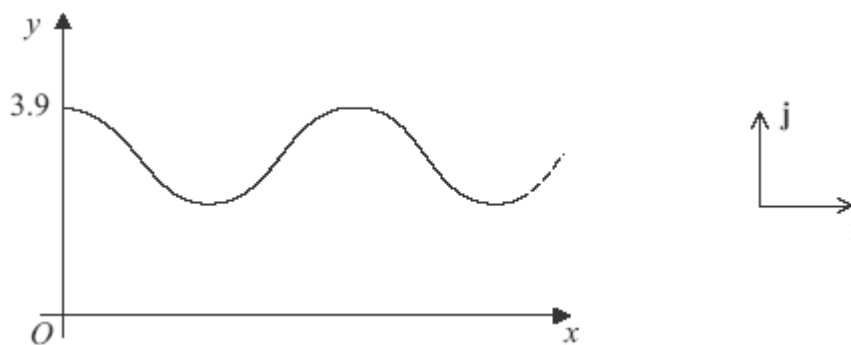
Q25.

A particle is initially at the origin, where it has velocity $(5\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. It moves with a constant acceleration $\mathbf{a} \text{ m s}^{-2}$ for 10 seconds to the point with position vector $75\mathbf{i}$ metres.

- (a) Show that $\mathbf{a} = 0.5\mathbf{i} + 0.4\mathbf{j}$. (3)
 - (b) Find the position vector of the particle 8 seconds after it has left the origin. (3)
 - (c) Find the position vector of the particle when it is travelling parallel to the unit vector $\mathbf{i}$. (6)
- (Total 12 marks)**

Q26.

Jane is on a ride in a theme park. Part of the curved path of the ride is shown in the diagram.



Jane's position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = 1.2t \mathbf{i} + (3 + 0.9 \cos t) \mathbf{j}$$

where the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed horizontally and vertically upwards respectively.

- (a) Find an expression for Jane's velocity at time t . (2)
 - (b) (i) Find an expression for Jane's speed at time t . (2)
 - (ii) Find Jane's maximum speed during the ride. (2)
- (Total 6 marks)

Q27.

A boat is initially at the origin, heading due east at 5 m s^{-1} . It then experiences a constant acceleration of $(-0.2\mathbf{i} + 0.25\mathbf{j}) \text{ m s}^{-2}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the initial velocity of the boat as a vector. (1)
 - (b) Find an expression for the velocity of the boat t seconds after it has started to accelerate. (2)
 - (c) Find the value of t when the boat is travelling due north. (3)
 - (d) Find the bearing of the boat from the origin when the boat is travelling due north. (6)
- (Total 12 marks)

Q28.

A particle P moves with acceleration $(-3\mathbf{i} + 12\mathbf{j}) \text{ m s}^{-2}$. Initially the velocity of P is $4\mathbf{i} \text{ m s}^{-1}$.

- (a) Find the velocity of P at time t seconds.

(2)

- (b) Find the speed of P when $t = 0.5$.

(3)

(Total 5 marks)

Q29.

A particle moves such that at time t seconds its acceleration is given by

$$(2 \cos t\mathbf{i} - 5 \sin t\mathbf{j}) \text{ m s}^{-2}$$

- (a) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 0$.

(3)

- (b) When $t = 0$, the velocity of the particle is $(2\mathbf{i} + 10\mathbf{j}) \text{ m s}^{-1}$.

Find an expression for the velocity of the particle at time t .

(5)

(Total 8 marks)

Q30.

The points A and B have position vectors $(3\mathbf{i} + 2\mathbf{j})$ metres and $(6\mathbf{i} - 4\mathbf{j})$ metres respectively. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in a horizontal plane.

- (a) A particle moves from A to B with constant velocity $(\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. Calculate the time that the particle takes to move from A to B .

(3)

- (b) The particle then moves from B to a point C with a constant acceleration of $2\mathbf{j} \text{ m s}^{-2}$. It takes 4 seconds to move from B to C .

- (i) Find the position vector of C .

(4)

- (ii) Find the distance AC .

(2)

(Total 9 marks)

Q31.

A particle moves in a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = (2t^3 - t^2 + 6)\mathbf{i} + (8 - 4t^3 + t)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(3)

- (b) (i) Find the velocity of the particle when $t = \frac{1}{3}$.

(2)

- (ii) State the direction in which the particle is travelling at this time.

(1)

- (c) Find the acceleration of the particle when $t = 4$.

(3)

- (d) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 4$.

(3)

(Total 12 marks)

Q32.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A yacht moves with a constant acceleration. At time t seconds the position vector of the yacht is $\mathbf{r}$ metres. When $t = 0$ the velocity of the yacht is $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$, and when $t = 10$ the velocity of the yacht is $(-\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$.

- (a) Find the acceleration of the yacht.

(3)

- (b) When $t = 0$ the yacht is 20 metres due east of the origin. Find an expression for $\mathbf{r}$ in terms of t .

(3)

- (c) (i) Show that when $t = 20$ the yacht is due north of the origin.

(2)

- (ii) Find the speed of the yacht when $t = 20$.

(4)

(Total 12 marks)

Q33.

A particle moves on a smooth horizontal surface with acceleration $(3\mathbf{i} - 5\mathbf{j}) \text{ m s}^{-2}$. Initially the velocity of the particle is $4\mathbf{j} \text{ m s}^{-1}$.

- (a) Find an expression for the velocity of the particle at time t seconds.

(2)

- (b) Find the time when the particle is travelling in the $\mathbf{i}$ direction.

(2)

- (c) Show that when $t = 4$ the speed of the particle is 20 m s^{-1} .

(4)

(Total 8 marks)

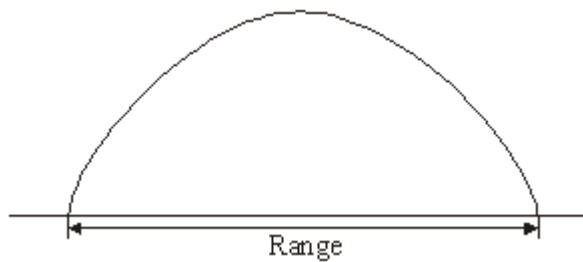
Q34.

A ball is hit from horizontal ground with velocity $(10\mathbf{i} + 24.5\mathbf{j}) \text{ m s}^{-1}$ where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertically upwards respectively.

- (a) State two assumptions that you should make about the ball in order to make predictions about its motion.

(2)

- (b) The path of the ball is shown in the diagram.



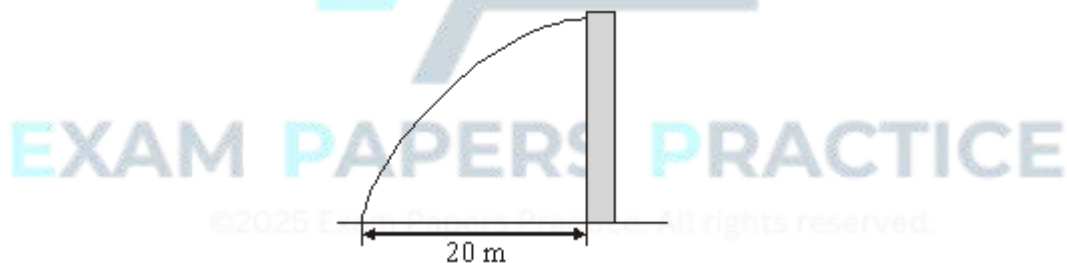
- (i) Show that the time of flight of the ball is 5 seconds.

(4)

- (ii) Find the range of the ball.

(2)

- (c) In fact the ball hits a vertical wall that is 20 metres from the initial position of the ball.



Find the height of the ball when it hits the wall.

(4)

- (d) If a heavier ball were projected in the same way, would your answers to part (b) of this question change? Explain why.

(2)

(Total 14 marks)

Q35.

A particle P , of mass 0.5 kg , moves under the action of a force $\mathbf{F}$ newtons, where $\mathbf{F} = 2t\mathbf{i} + 1.5\mathbf{j}$ at time t seconds.

- (a) Find the acceleration of P at time t .

(1)

- (b) The particle P starts its motion from rest.

Find the velocity of P at time t .

(2)

- (c) The particle P starts its motion at the point O .

(i) Find the position vector of P relative to O at time t .

(3)

(ii) Find the distance of P from O when it is moving parallel to the vector $\mathbf{i} + \mathbf{j}$.

(5)

(Total 11 marks)



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